

3(a)(i)	row numbers refer to Figure 3.1 any three from: - both DNA and RNA have purines, which are adenine and guanine (row 1) ; - both DNA and RNA have the pyrimidine cytosine but DNA has thymine and RNA has uracil (row 2) ; - in DNA, purine pairs with pyrimidine / adenine pairs with thymine and guanine pairs with cytosine B; A A-T and G-C A single ring base-paired with double ring base <i>instead of purine - pyrimidine</i> - RNA is formed by base pairing during transcription from (template strand of) DNA so can only have 4 bases ; if no marks gained allow one mark for naming all five bases / letters in the correct context(s)	3
3(a)(ii)	any two from: base joins with a pentose sugar, which joins with a phosphate (group) ; A (base) joins with pentose sugar and phosphate (group) A deoxyribose (for DNA) / ribose (for RNA) to form a nucleotide ; AVP ; e.g. correct <i>ref. to</i> nucleoside (i.e. sugar and base) two further phosphate groups added / become activated nucleotide join to adjacent nucleotide by phosphodiester bond	2
3(a)(iii)	any four from: - each amino acid is coded by a, triplet of bases / triplet of nucleotides / group of three bases / group of three nucleotides / a codon ; (reading frame is) non-overlapping ; - no gaps / <i>idea of</i> continuous sequence ; <i>ref. to</i> start codon and stop codon ; - sequence of, codons / triplets (of bases), determines the <u>primary structure of polypeptide</u> ; - AVP ; } e.g. <i>ref. to</i> degenerate / redundant, code or more than one codon for some amino acids - AVP ; } start codon, is AUG / codes for met stop codon / UGA / UAG / UAA, do not coding for an(y) amino acid / for chain termination	4

Question	Answer	Marks										
3(b)(i)	<i>if a row contains more than one process both or all must come from this list below</i> <table><tr><th>stage</th><th>process involving DNA in the nucleus</th></tr><tr><td>1</td><td>transcription / RNA synthesis / RNA production / error checking / unwinding / uncoiling</td></tr><tr><td>2</td><td>(semi-conservative) replication / sister chromatid formation / forming complex with histones / synthesis of histones</td></tr><tr><td>3</td><td>error checking</td></tr><tr><td>4</td><td>condensation / coiling / supercoiling / described in terms of histones / shortens / thickens / AW I becomes visible</td></tr></table>	stage	process involving DNA in the nucleus	1	transcription / RNA synthesis / RNA production / error checking / unwinding / uncoiling	2	(semi-conservative) replication / sister chromatid formation / forming complex with histones / synthesis of histones	3	error checking	4	condensation / coiling / supercoiling / described in terms of histones / shortens / thickens / AW I becomes visible	3
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3(b)(ii)	<i>any two from:</i> I any points to do with cytokinesis in plant cells cleavage furrow forms ; constriction of cytoplasm / cell (surface) membrane ‘pinches in’ ; membrane fusion occurs ; organelles shared between daughter cells ; AVP ; e.g. ring of microtubules formed underneath cell (surface) membrane (pulls in membrane)	2										

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3(b)(iii)	<i>award mp1 for either enzyme</i> 1 forms phosphodiester bonds or (function) during, DNA replication / S phase (of interphase) ; <i>DNA polymerase</i> 2 forms bonds between / joins, activated / free, nucleotides, and, elongating / AW, strand or in a 5' to 3' direction ; A lengthening / growing / new / synthesising <i>for elongating</i> I ref. to template strand 3 error checking / proof reading ; <i>ligase</i> 4 forms bonds between / joins, Okazaki fragments / DNA sections on the lagging strand ;	3